The derivative of u with respect to x when x varies and y remains constant is called the **partial derivative of u with respect to x,** denoted by $\frac{\partial u}{\partial x}$ (or) u_x and is defined as

$$\frac{\partial u}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}$$

provided the limit exists. Here Δx is a small change in x

The derivative of u with respect to y, when y varies and x remains constant is called the **partial derivative of u with respect to y,** denoted by $\frac{\partial u}{\partial y}$ (or) u_y and is defined as

$$\frac{\partial u}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y+\Delta y) - f(x,y)}{\Delta y}$$

provided the limit exists. Here Δy is a small change in y.

$\frac{\partial u}{\partial x}$ is also written as $\frac{\partial}{\partial x} f(x,y)$ (or) $\frac{\partial f}{\partial x}$.

Similarly $\frac{\partial u}{\partial y}$ is also written as $\frac{\partial}{\partial y} f(x,y)$ (or) $\frac{\partial f}{\partial y}$

The process of finding a partial derivative is called **partial differentiation.**

6.4.1 Successive partial derivatives

Consider the function $u = f(x, y)$. From this we can find $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$. If $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ are functions of x and y, then they may be differentiated partially again with respect to either of the independent variables, (x or y) denoted by $\frac{\partial^2 u}{\partial y^2}$, $\frac{\partial^2 u}{\partial x^2}$ $\frac{\partial^2 u}{\partial y \partial x}$, $\frac{\partial^2 u}{\partial y \partial x}$.

These derivatives are called second order partial derivatives. Similarly, we can find the third order partial derivatives, fourth order partial derivatives etc., if they exist. The process of finding such partial derivatives are called **successive partial derivatives.**

If we differentiate $u = f(x,y)$ partially with respect to x and again differentiating partially with respect to y,

we obtain $\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)$ i.e., $\frac{\partial^2 u}{\partial y \partial x}$

Similarly, if we differentiate $u = f(x,y)$ partially with respect to y and again differentiating partially with respect to x, we obtain $\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)$ i.e., $\frac{\partial^2 u}{\partial x \partial y}$

NOTE

If $u(x,y)$ is a continuous function of x and y, then $\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$.

Homogeneous functions

A function $f(x,y)$ of two independent variables x and y is said to be homogeneous in x and y of degree n if $f(tx,ty) = t^n f(x,y)$ for $t > 0$.

6.4.2 Euler's theorem and its applications

Euler's theorem for two variables:

If $u = f(x,y)$ is a homogeneous function of degree n, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$

Example 6.35

If $u = x^2(y-x) + y^2(x-y)$, then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = -2(x-y)^2$.

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